WORKED SOLUTION

STEP III 2001, Question 8

Pure • Complex Loci

STEP III 2001, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q8

The key idea

Write $z = x + \mathrm{i}y$ and expand: both equations in (i) collapse to the same circle. For the argument condition in (ii), compute $\frac{z-2}{z}$ explicitly; an argument of $\frac{\pi}{4}$ means "real part equals imaginary part, and both are positive", which is the circle again plus an inequality. For (iii), substitute $z = 2/w$ into whichever form of the circle makes the algebra tidiest. The second form in (i) is there for exactly this purpose.

Solution

Part (i)

Let $z = x + \mathrm{i}y$. The first equation is

$$(x-1)^2 + (y-1)^2 = 2 \iff x^2 + y^2 - 2x - 2y = 0.$$

The second is

$$\begin{aligned} & (x-1)^2 + (y+1)^2 = 2[(x-1)^2 + y^2] \\ \iff\ & x^2 + y^2 - 2x + 2y + 2 = 2x^2 + 2y^2 - 4x + 2 \\ \iff\ & 0 = x^2 + y^2 - 2x - 2y. \end{aligned}$$

Each step is reversible, so both equations are equivalent to $x^2 + y^2 - 2x - 2y = 0$ and describe the same locus. That locus is the circle with centre $1 + \mathrm{i}$ and radius $\sqrt{2}$. It passes through 0, 2 and 2i (and $2 + 2\mathrm{i}$), and its lowest point is $1 + (1 - \sqrt{2})\mathrm{i}$, just below the real axis.

Part (ii)

For $z \neq 0$,

$$\frac{z-2}{z} = \frac{(z-2)\bar{z}}{|z|^2} = \frac{x^2 + y^2 - 2x + 2\mathrm{i}y}{x^2 + y^2},$$

since $(z-2)\bar{z} = |z|^2 - 2\bar{z} = x^2 + y^2 - 2x + 2\mathrm{i}y$.

Ask yourself What exactly does arg(something) $= \frac{\pi}{4}$ say about its real and imaginary parts?

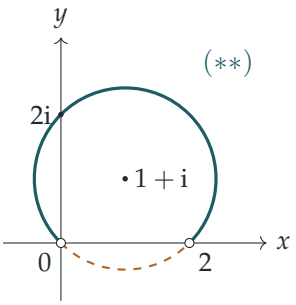
A nonzero complex number $a + \mathrm{i}b$ has argument $\frac{\pi}{4}$ exactly when it lies on the ray $a = b > 0$. Equality alone isn't enough: $a = b < 0$ gives argument $-\frac{3\pi}{4}$. The denominator $x^2 + y^2$ is positive, so $(**)$ holds if and only if

$$x^2 + y^2 - 2x = 2y \quad \text{and} \quad 2y > 0,$$

that is, if and only if z lies on the circle from (i) with $y > 0$. The condition $y > 0$ automatically rules out $z = 0$, where the quotient is undefined, and $z = 2$, where it is zero and has no argument.

So $(**)$ describes the part of the circle above the real axis: the major arc from 0 round through 2i and $2 + 2\mathrm{i}$ to 2, with both endpoints excluded. The short arc below the real axis gives argument $-\frac{3\pi}{4}$ instead.

This fits the geometry. $\arg\frac{z-2}{z}$ is the angle at z between the chords to 0 and to 2. The chord from 0 to 2 subtends a right angle at the centre $1 + \mathrm{i}$, so by the angle-at-the-centre theorem it subtends $\frac{\pi}{4}$ at every point of the major arc.



The whole circle is $(*)$; the solid arc (endpoints 0 and 2 excluded) is $(**)$, and the dashed arc is not part of $(**)$.

Part (iii)

Since $w = 2/z$ we have $z \neq 0$ and $z = 2/w$. Use the second form of the circle, $|z - (1-\mathrm{i})|^2 = 2|z-1|^2$. Substituting,

$$z - 1 = \frac{2-w}{w}, \qquad z - (1-\mathrm{i}) = \frac{2-(1-\mathrm{i})w}{w} = \frac{(1-\mathrm{i})[(1+\mathrm{i})-w]}{w},$$

using $\frac{2}{1-\mathrm{i}} = 1 + \mathrm{i}$. Since $|1-\mathrm{i}|^2 = 2$, the equation becomes

$$\frac{2\,|w-(1+\mathrm{i})|^2}{|w|^2} = \frac{2\,|w-2|^2}{|w|^2} \iff |w-(1+\mathrm{i})| = |w-2|.$$

So w is equidistant from $1+\mathrm{i}$ and 2: the locus is the perpendicular bisector of the segment joining them. The midpoint is $\frac{3}{2}+\frac{1}{2}\mathrm{i}$ and the segment has direction $1-\mathrm{i}$, so the bisector has direction $1+\mathrm{i}$. With $w=u+\mathrm{i}v$, the locus is the line

$$v=u-1.$$

Every point of this line arises. Each w on it is nonzero, since 0 isn't on the line, and $z=2/w$ then satisfies $(*)$ by the reversible steps above. The only point of the circle with no image is $z=0$, which is "sent to infinity". A circle through the origin maps to a line under $w=2/z$, as expected.

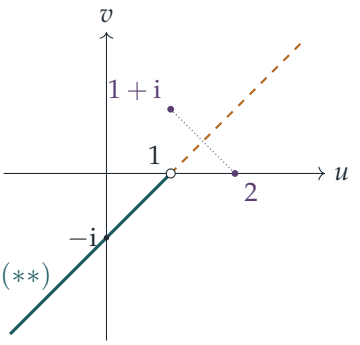
For the part corresponding to $(**)$, notice that

$$\frac{z-2}{z}=1-\frac{2}{z}=1-w,$$

so $(**)$ becomes $\arg(1-w)=\frac{\pi}{4}$, i.e. $1-w=r\,\mathrm{e}^{\mathrm{i}\pi/4}$ with $r>0$. Hence

$$w=1-\frac{r}{\sqrt{2}}(1+\mathrm{i}),\qquad r>0,$$

which is the half-line of $v=u-1$ starting at $w=1$ (excluded) and running down to the left: the points with $u<1$, equivalently $v<0$. This matches (ii): $\operatorname{Im} w=-2y/|z|^2$, so $y>0$ corresponds to $v<0$. For example, $z=2\mathrm{i}$ maps to $w=-\mathrm{i}$, and $z=2$ maps to the excluded endpoint $w=1$.



The whole line $v=u-1$ is the image of $(*)$; the solid half-line ($u<1$, with $w=1$ excluded) is the image of $(**)$. The dotted segment joins the two points that the line bisects.

Checks

- $z=2\mathrm{i}$ is on the circle and above the axis: $\frac{2\mathrm{i}-2}{2\mathrm{i}}=1+\mathrm{i}$, which has argument $\frac{\pi}{4}$. Its image $w=-\mathrm{i}$ satisfies $v=u-1$ with $u<1$.
- The top of the circle, $z=1+(1+\sqrt{2})\mathrm{i}$, maps to $w\approx 0.2929-0.7071\mathrm{i}$, and indeed $0.2929-1=-0.7071$.
- A point below the axis, $z=1+(1-\sqrt{2})\mathrm{i}$, gives $\arg\frac{z-2}{z}=-\frac{3\pi}{4}$. Its image has $u>1$, on the other half of the line.

Common slips

- Treating $\arg Z = \frac{\pi}{4}$ as just $\operatorname{Re} Z = \operatorname{Im} Z$, and so claiming the whole circle. You also need both parts positive.
- Forgetting to exclude $z = 0$ and $z = 2$ in (ii), or $w = 1$ in (iii).
- Expecting the image of a circle to be another circle. Under $w = 2/z$, circles through the origin go to straight lines.

Beyond STEP

The map $w = 2/z$ is a Möbius transformation. These maps send circles and lines to circles and lines, and a circle goes to a line exactly when it passes through the point sent to infinity. Here the endpoints 0 and 2 of the arc go to ∞ and 1. The quotient $\frac{z-2}{z-0}$, which measures the angle subtended by those endpoints, becomes $1 - w$: the direction from the image point to 1. That is why the arc in (ii) becomes a half-line in (iii). Needham's *Visual Complex Analysis* develops this geometric viewpoint beautifully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.